APM1220 FORMATIVE ASSESSMENT 5

Principal Component Analysis of the World Happiness Report 2021

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Software: Python (Google Colab) **Observations:** 149

Objective

This assessment investigates whether seven quantitative measures of cross-country well-being can be summarized using fewer principal components. Variables are standardized and PCA is based on the correlation matrix, following the assessment instructions and the course module on Principal Components.

Dataset and variables

The dataset contains 149 country observations and 20 original columns. The analysis uses Ladder score, Logged GDP per capita, Social support, Healthy life expectancy, Freedom to make life choices, Generosity, and Perceptions of corruption. Country name and Regional indicator are excluded from PCA. The seven selected variables contain no missing values.

Part A — Data Preparation and Descriptive Analysis

A1. Number of observations and variables

There are 149 observations and seven quantitative variables selected for PCA.

A2. Mean and standard deviation

Variable	Mean	Standard deviation
Ladder score	5.533	1.074
Logged GDP per capita	9.432	1.159
Social support	0.815	0.115
Healthy life expectancy	64.993	6.762
Freedom to make life choices	0.792	0.113
Generosity	-0.015	0.151
Perceptions of corruption	0.727	0.179

A3. Why standardize the variables?

The variables are measured on different numerical scales. Standardization gives each variable a mean of zero and standard deviation of one, so variables with larger numerical variances do not dominate PCA simply because of their units. PCA is performed using the correlation matrix, as required.

A4. Correlation matrix

Variable	Ladder	GDP	Support	Health	Freedom	Generosity	Corruption
Ladder	1.000	0.790	0.757	0.768	0.608	-0.018	-0.421
GDP	0.790	1.000	0.785	0.859	0.432	-0.199	-0.342
Support	0.757	0.785	1.000	0.723	0.483	-0.115	-0.203
Health	0.768	0.859	0.723	1.000	0.461	-0.162	-0.364
Freedom	0.608	0.432	0.483	0.461	1.000	0.169	-0.401
Generosity	-0.018	-0.199	-0.115	-0.162	0.169	1.000	-0.164

Variable	Ladder	GDP	Support	Health	Freedom	Generosity	Corruption
Corruption	-0.421	-0.342	-0.203	-0.364	-0.401	-0.164	1.000

Logged GDP per capita and Healthy Life Expectancy have a strong positive correlation (r = 0.859). Ladder Score is positively correlated with Logged GDP per capita (r = 0.790) and Social Support (r = 0.757). Ladder Score and Perceptions of Corruption have a negative correlation (r = -0.421), while Ladder Score and Generosity have almost no linear relationship (r = -0.018). These associations suggest overlapping variation among some measures of well-being; correlation does not establish causation.

Part B — Principal Component Analysis

B1–B4. Eigenvalues and variance explained

Component	Eigenvalue	Proportion	Cumulative proportion
PC1	3.914	0.559	0.559
PC2	1.289	0.184	0.743
PC3	0.708	0.101	0.844
PC4	0.518	0.074	0.918
PC5	0.251	0.036	0.954
PC6	0.193	0.028	0.982
PC7	0.126	0.018	1.000

PC1 explains 55.9% of total variance, and PC2 explains 18.4%. Together, they explain 74.3% of the total variance. The sum of the seven eigenvalues is 7.000, as expected for a correlation matrix of seven standardized variables.

B5. Eigenvectors / component coefficients

Variable	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Ladder score	-0.464	0.040	-0.088	-0.013	0.095	0.869	0.105
Logged GDP per capita	-0.459	-0.197	0.002	-0.256	-0.174	-0.124	-0.800
Social support	-0.433	-0.163	-0.324	-0.087	0.704	-0.361	0.218
Healthy life expectancy	-0.453	-0.146	0.032	-0.230	-0.612	-0.234	0.539
Freedom to make life choices	-0.341	0.365	-0.167	0.811	-0.163	-0.174	-0.092
Generosity	0.042	0.753	-0.466	-0.449	-0.083	-0.051	-0.041
Perceptions of corruption	0.252	-0.460	-0.801	0.119	-0.238	0.107	-0.037

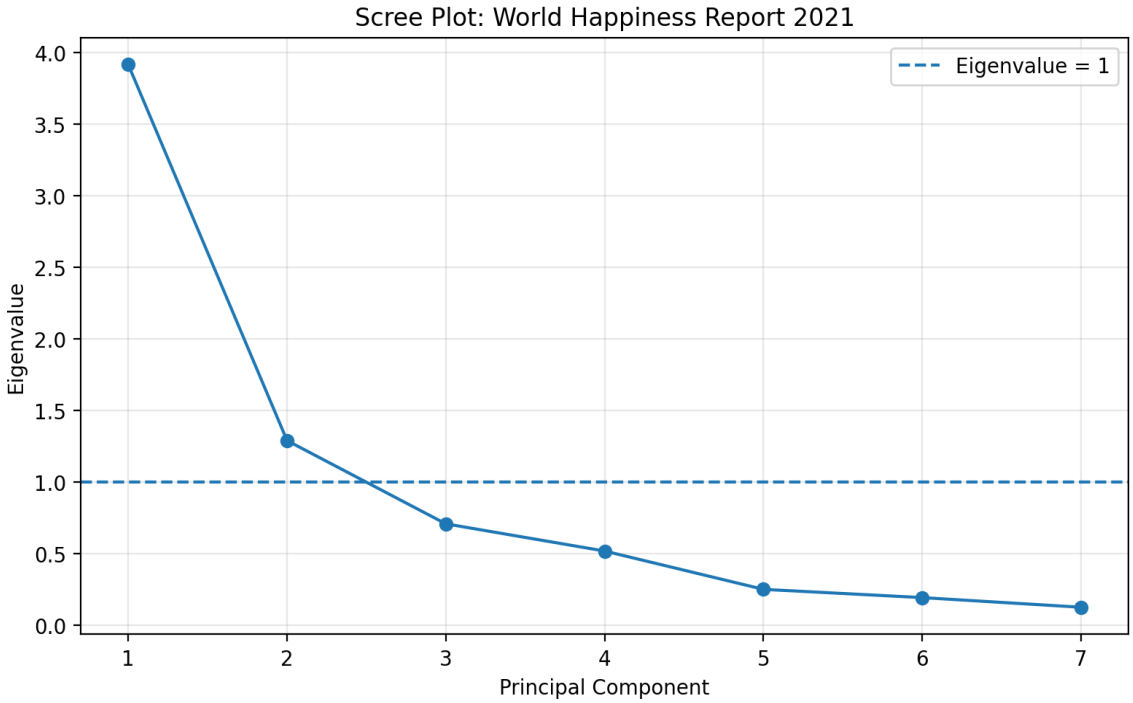
B6. Largest absolute coefficients in PC1

Variable	PC1 coefficient	Absolute coefficient
Ladder score	-0.464	0.464
Logged GDP per capita	-0.459	0.459
Healthy life expectancy	-0.453	0.453
Social support	-0.433	0.433
Freedom to make life choices	-0.341	0.341
Perceptions of corruption	0.252	0.252
Generosity	0.042	0.042

PC1 has the largest absolute coefficients for Ladder Score, Logged GDP per capita, Healthy Life Expectancy, and Social Support, followed by Freedom to Make Life Choices. Generosity contributes little. Eigenvector signs are arbitrary and may all be reversed without changing the PCA solution.

Part C — Determining the Number of Components

C1. Scree plot



C2. Apparent elbow

The scree plot shows a sharp decrease from PC1 to PC2, another decrease from PC2 to PC3, and a more gradual decline after PC3. The apparent elbow is around PC2–PC3, though visual identification is somewhat subjective.

C3. Kaiser criterion

The Kaiser criterion retains components with eigenvalues greater than 1. PC1 (3.914) and PC2 (1.289) meet this criterion; PC3 through PC7 have eigenvalues below 1. Thus, the Kaiser criterion recommends two components.

C4. Recommended number of components

Two components are recommended. PC1 and PC2 explain 74.3% of total variance and both have eigenvalues greater than 1. Three components would explain 84.4%, but PC3 has an eigenvalue of 0.708. The two-component solution offers a reasonable balance between data reduction and retained variation.

C5. Why multiple criteria matter

No single criterion gives a definitive answer. The Kaiser criterion provides a numerical rule, the scree plot shows where eigenvalues level off, and cumulative variance measures how much variation is retained. Subject-matter interpretation also matters.

Part D — Interpretation of Principal Components

D1. Principal Component 1

PC1 has large absolute coefficients for Ladder Score (-0.464), Logged GDP per capita (-0.459), Healthy Life Expectancy (-0.453), Social Support (-0.433), and Freedom to Make Life Choices (-0.341). Perceptions of Corruption has a coefficient of 0.252, while Generosity has a small coefficient of 0.042. This pattern indicates that PC1 summarizes shared variation in life evaluation, economic prosperity, health, social support, and freedom. The negative signs do not mean these characteristics are undesirable; an eigenvector's signs can be reversed without changing the solution.

D2. Principal Component 2

PC2 is dominated by Generosity (0.753), followed by Perceptions of Corruption (-0.460) and Freedom to Make Life Choices (0.365). It represents a secondary dimension contrasting generosity and freedom with the reported corruption-perception measure. These coefficients describe statistical contrast, not causation.

D3. Descriptive label for PC1

A suitable label for PC1 is **General Well-Being and Socioeconomic Conditions**, because its largest coefficients are associated with life evaluation, economic prosperity, health, social support, and freedom.

D4. Original variables represented

The two retained components summarize all seven original quantitative variables, preserving 74.3% of their total variance and reducing the representation from seven variables to two components.

D5. Conclusion

The PCA suggests that seven measures of cross-country well-being can be summarized by two principal components that explain 74.3% of total variance. PC1 represents general well-being and socioeconomic conditions, while PC2 captures a secondary contrast characterized mainly by generosity, freedom, and perceptions of corruption. PCA reduces dimensionality but does not establish causal effects.

Reproducibility and source

The accompanying Python script reproduces the descriptive statistics, correlation matrix, PCA eigenvalues, variance proportions, eigenvectors, and scree plot. Dataset mirror: https://huggingface.co/datasets/jason1966/ajaypalsinghlo_world-happiness-report-2021. Cite the original World Happiness Report 2021 as required by the assessment.